Mohsen Asghari Ilani

Data Scientist | Researcher | Machine Learning Engineer

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Profile

Data scientist and researcher with an engineering background and a focus on machine learning for socio-economic, energy and environmental risk. Experienced in building and deploying predictive models, from deep learning for imaging to forecasting household energy vulnerability, with peer-reviewed work in *Risk Analysis*, *International Journal of Forecasting*, *Scientific Reports* and *IEEE Geoscience and Remote Sensing Letters*. Currently a PhD researcher at Swansea University funded by ESRC/UKRI, and Senior Data Scientist at Dotin.

Education

MSc in Social Research – Distinction — **Sep 2025 – Dec 2026**
Swansea University — Swansea, UK
Funded by WGSSS | ESRC | UKRI

MSc in Mechanical Engineering — **Sep 2016 – Feb 2019**
University of Tehran — Tehran, Iran | GPA 3.4

BSc in Mechanical Engineering — **Sep 2009 – Jul 2013**
Islamic Azad University — Iran | GPA 3.6

Professional Experience

Senior Data Scientist — **Jul 2026 – Present**
Dotin

- Develop and deploy machine learning models for real-world applications.
- Analyse large datasets to extract actionable insights for decision-making.
- Collaborate with cross-functional teams to deliver data-driven solutions.

Independent Researcher — **Feb 2019 – Present**
Machine Learning and Socio-Economic Analytics

- Conduct research in machine learning for socio-economic, energy, environmental and medical applications.
- Publish in high-impact peer-reviewed journals (see Publications).
- Contribute to collaborative academic and industry research projects.

Mechanical Engineer — **Aug 2013 – Jul 2016**
Iran Khodro Agency — Iran

- Contributed to research and development projects.
- Worked with technical teams to deliver data-driven engineering solutions.

Selected Publications

[1] **Who Becomes Fuel Vulnerable, and When? Early Warning of Household Vulnerability Risk in the UK**
M. Asghari Ilani, H. Eskandari, M. Banad
Risk Analysis (2026), accepted.

[2] **Forecasting Groundwater Irrigation Risk under Salinity–Sodicity Stress Using Hydrochemical–Climate Data Fusion: A Tabular, Spatial, and Ordinal Learning Framework for Environmental Management**
M. Asghari Ilani, S. Mokhtar Tehran, A. Kavei, M. Banad, H. Eskandari
Journal of Environmental Management (2026).

[3] **Do Electricity-Grid Signals Improve Quarterly Forecasts of UK Greenhouse-Gas Emissions? A Layered Audit of Apparent Machine-Learning Gains**
M. Asghari Ilani, H. Eskandari, Y. M. Banad
Energy and AI (2026).

[4] **TT-Weighted MRI-Based Brain Tumor Classification Using Hybrid Deep Learning Models**
M. Asghari Ilani, D. Shi, Y. M. Banad
Scientific Reports 15(1), 1010 (2025).

[5] **Enhanced HS-SNet: Improved SAR Image Flood Segmentation with Learnable Histogram Layers and Active Contour Model**

M. Asadi, S. Saeidi, M. Kordani, **M. Asghari Ilani**, Y. M. Banad
IEEE Geoscience and Remote Sensing Letters 22, 1–5 (2025).

[6] **Comparative Study of Long Short-Term Memory (LSTM), Bidirectional LSTM, and Traditional Machine Learning Approaches for Energy Consumption Prediction**
H. Alizadegan, R. Rashidi, **M. Asghari Ilani**, M. Banad
Energy Exploration & Exploitation 43(1), 281–301 (2025).

Awards & Achievements

1st Place – RoboCup 2025, Rescue Agent Simulation League **15–27 Jul 2025**
RoboCup 2025, Brazil
Team: M. A. Ghaemi, H. Alizadegan, M. Asghari Ilani

Rescue Agent Simulation Competition
Khatam University Tehran, Iran

Technical Skills

Machine Learning	Deep learning, computer vision, natural language processing, time-series forecasting
Data Science	Statistical modelling, predictive analytics, computational modelling, algorithm design
Programming	Python, R, MATLAB, SQL
Tools & Frameworks	TensorFlow, PyTorch, scikit-learn, Pandas, Docker
Languages	English (professional), Persian/Farsi (native)

References

Mike Banad
University of Oklahoma, USA
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